DELHI PUBLIC SCHOOL JANKIPURAM

SET–I

HALF YEARLY EXAMINATION: (2024-25)

SUBJECT: SCIENCE (086)

GRADE: IX

Time – 3 hrs. **M.M. 80**

Name- _____ **Roll No-** _____ **Date-** _____

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General Instructions:

Read the following instructions carefully and follow them:

- This question paper consists of 39 questions in 5 sections
- All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.
- Section A consists of 20 objective type questions carrying 1 mark each
- Section B consists of 6 Very Short questions carrying 02 marks each.
- Section C consists of 7 Short Answer type questions carrying 03 marks each.
- Section D consists of 3 Long Answer type questions carrying 05 marks each.
- Section E consists of 3 source-based/case-based units of assessment of 04 marks each with sub-parts.

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SECTION A

CHOOSE THE CORRECT OPTION: (1*20 = 20)

Q1) Dry ice is

a) water in solid state.
b) water in gaseous state
c) CO_2 in liquid state.
d) CO_2 in solid state

Q2) Which of these is an alloy?

a) silver.
b) copper
c) bronze.
d) aluminium

Q3) Which of the following is a homogeneous mixture?

a) Sand.
b) Oil and water
c) salt and sand mixture.
d) Air

Q4) Cell organelle important for membrane biogenesis is

a) Endoplasmic reticulum
b) Mitochondria.
c) Golgi apparatus.
d) Nucleus

Q5) The connective tissue that provide support to the internal tissue is

a) Blood
b) Areolar connective tissue
c) Adipose connective tissue
d) Cartilage

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Q6) Acceleration due to gravity of the Earth increases with
a) decrease in the height from the surface of the Earth
b) increase in the height from the surface of the Earth.
c) increase in the depth from the surface of the Earth.
d) Increase in the temperature of the Earth

Q7) Which condition out of the following will increase the evaporation of water?
a) Increase in temperature of water
b) Decrease in temperature of water
c) Less exposed surface area of water
d) Adding common salt to water

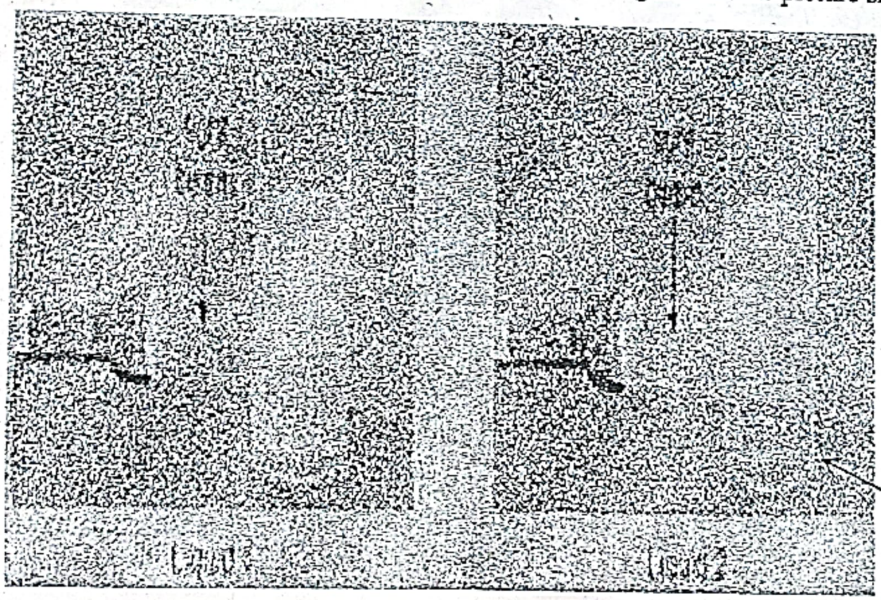
Q8) If a body is moving at constant speed in a circular path, its
a) Velocity is constant and its acceleration is zero
b) Velocity and acceleration are both changing in direction only
c) Velocity and acceleration are both increasing
d) Velocity is constant and acceleration is changing direction

Q9) The acceleration of a car that speeds up from 12 m/s to 30 m/s in 15 seconds is
a) 2.4 m/s^2
b) 1.2 m/s^2
c) 2 m/s^2
d) 5.2 m/s^2

Q10) The skin is made up of _____ epithelial
a) Ciliated
b) Columnar.
c) Squamous
d) Cuboidal

Q11) ____ is involved in the formation of lysosomes
a) Plastid
b) Golgi Apparatus
c) Chromosome
d) Endoplasmic reticulum

Q12) Tina passes light beams through two liquid mixtures in separate glasses. The picture shows Tina's observations.



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Based on the behavior of the light beam what are liquids 1 and 2?

	Liquid 1	Liquid 2
a	solution	colloid
b	colloid	suspension
c	solution	suspension
d	suspension	colloid

Q13) ____ plays crucial role in detoxifying poisons and drugs in vertebrates

a) Long Endoplasmic reticulum
b) Endoplasmic reticulum.
c) RER.
d) SER

Q14) The girth of root or stem increases due to

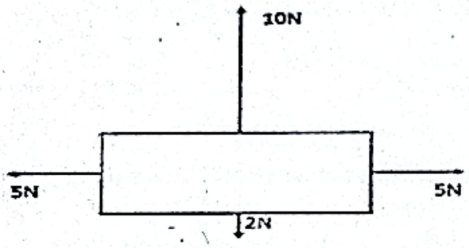
a) Apical meristem
b) Lateral meristem
c) Intercalary meristem
d) None of these

Q15) Which of the following can sometimes be 'zero' for a moving body?

i. Distance travelled
ii. Average velocity
iii. Average speed
iv. Displacement

a) Only (i)
b) (i) and (ii)
c) (ii) and (iv)
d) Only (iv)

Q16) What is the resultant force?



a) 8 N upwards
b) 12 N downwards
c) 12 N upwards
d) 8 N downwards

ASSERTION AND REASONING TYPE OF QUESTIONS :

DIRECTION: In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as:

a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, and R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Q17) **Assertion (A):** The ratio of the average velocity and average speed of a body is always greater than 1.
Reason(R): The average speed is always greater than average velocity.

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Q18) **Assertion:** The outer membrane of mitochondria is folded into cristae
Reason: Cristae increases the surface area a

Q19) **Assertion (A):** Pure substances have fixed melting and boiling points.
Reason (R): The particles in a pure substance are all of the same kind and have the same chemical properties.

a

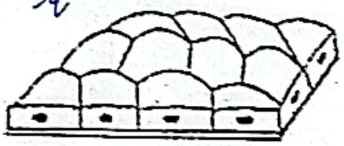
Q20) **Assertion:** Newton's third law applies to all types of forces. e.g. gravitational, electric or magnetic forces etc.
Reason: Newton's third law of motion is applicable only when bodies are in motion.

SECTION – B

VERY SHORT ANSWER TYPE OF QUESTIONS: (2*6 = 12)

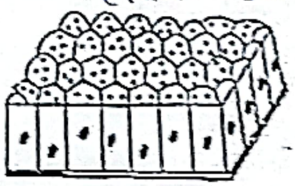
Q21) Identify and differentiate between the given diagrams A and B

Squamous



A

Ciliated columnar



B

Q22) a) Differentiate between the types of ER.
b) Give functions of ER.

Q23) State any two natural phenomena explained by universal law of Gravitation.

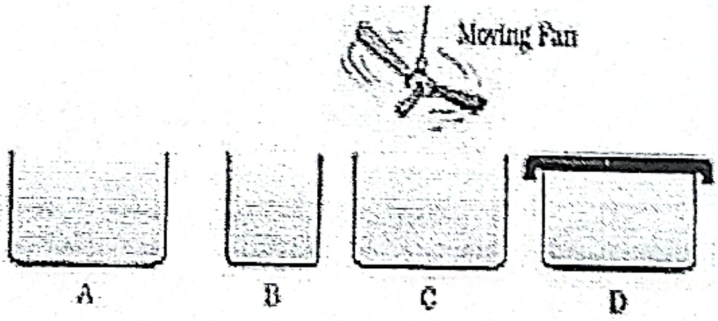
Q24) Where is fat stored in the body? How is it useful for the organisms?

OR

Draw a labeled diagram of neuron.

Q25) Mention the contrasting characters between the outermost covering in plant cells and animal cells.

Q26) Look at Fig. and suggest in which of the vessels A, B, C or D , the rate of evaporation will be the highest? Justify.



OR

Water as ice has a cooling effect, whereas water as steam may cause severe burns. Explain these observations.

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SECTION – C

SHORT ANSWER TYPE OF QUESTIONS: (3*7 = 21)

Q27) The gravitational force between two identical bodies is 200 N. Mass of both bodies becomes twice and separation becomes half. What will be a new force between them?

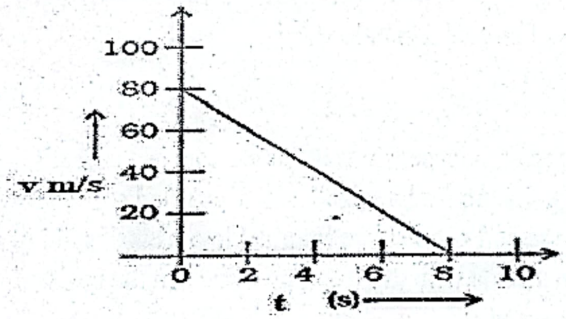
Q28) Can you describe your understanding of Vascular tissues in plants and explain its different types on the basis of their function.

Q29) a) Describe structure and function of cork in plants.
b) Draw a labeled diagram showing the location of meristematic tissue in plants.

Q30) During an experiment the students were asked to prepare a 10% (Mass/Mass) solution of sugar in water. Ramesh dissolved 10g of sugar in 100g of water while Sarika prepared it by dissolving 10g of sugar in water to make 100g of the solution.
(a) Are the two solutions of the same concentration?
(b) Compare the mass % of the two solutions.

Q31) Substance A has high compressibility and can be easily liquefied. It can take up the shape of any container. Predict the nature of the substance. Enlist four properties of this state of matter.

Q32) Velocity versus time graph of a ball of mass 0.1kg rolling on a concrete floor is shown below.



a) Name the physical quantity measured by the velocity time graph Distance in a unit time
b) Calculate the acceleration from the above graph. 10m/s²

OR

a) Derive the expression for velocity of a cyclist moving along the circular path with uniform speed.
b) Calculate the distance travelled when the body moving at 5m/s changes its velocity to 10m/s with the acceleration of 5m/s?

Q33) a) A car travels 12 m north, then 5 m east. What is the distance covered and the displacement of the car? State whether these quantities are scalar or vector.
b) A ball is thrown up with a speed of 15ms^{-1}. How high will it go before it begins to fall? (g= 9.8ms^{-2})

SECTION - D

LONG ANSWER TYPE OF QUESTIONS: (5*3 = 15)

Q34) a) Blood is a type of connective tissue. Name the matrix of blood.
b) Write the composition of blood.

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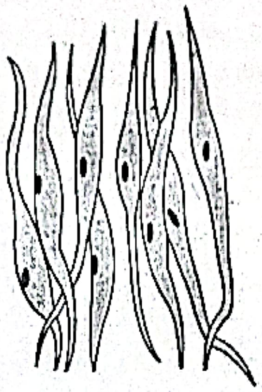
c) Mention any two function of blood as a connective tissue.

OR

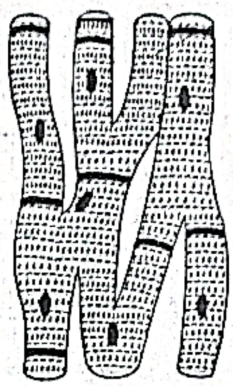
a) Identify the muscle in the below diagram and differentiate between them on the basis of structure.



A



B



C

b) What is the name of bone cells? Describe the function of bones.

Q35) Rohan conducted an experiment where he mixed some milk with water. He observed that the mixture appeared uniform, but when he passed a beam of light through it, the light scattered. Based on your understanding of colloidal solutions, answer the following questions:

1. Identify the type of mixture formed when Rohan mixed milk with water. Explain your reasoning.
2. Describe the Tyndall effect observed in the experiment and explain why it occurs in colloidal solutions.
3. List two everyday examples of colloidal solutions.

OR

a) How are Sol, Solutions and Suspensions different from each other?

b) During a science experiment, Aisha mixed iron filings and sulfur powder in a beaker. She then heated the mixture and noticed that it turned into a black solid of iron sulphide. Compare the properties of iron sulfide with those of iron filings on the basis of appearance and magnetic property.

Q36) a) Identify the law of motion that governs the recoiling of the gun when it is fired. Also mention the quantity that is conserved during the firing process.

b) Using Newton's law of motion, derive the relation between force and acceleration.

c) Two objects A and B, having mass 100 kg and 75 kg, moving with velocity 40 km/h and 6 km/h respectively.

Answer the following:

i. Which will have greater inertia? explain it.

ii. Which will have greater momentum? Give reasons.

OR

a) Name the physical quantity that measures inertia. Give its SI unit.

b) It is necessary to run along with the moving bus in the same direction of the bus, while alighting from the bus. Give reasons.

c) A car starts from rest and rolls down a hill with constant acceleration. It travels a distance of 800 m in 10 seconds. Find its acceleration. Also, find the force acting on it if its mass is 800 kg.

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